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Case Report

Congenital Absence of the Right Coronary Artery with an Ultra-Dominant Circumflex Artery: Multimodality Imaging Diagnosis

Louis Fernando Robles-Fernandes [a,*], *Erick Rey Palma-Fierro* [a], *Mariana Cáñez-Hernández* [a], *Axel Hiram Hernández-Pineda* [a], *Daniel Campuzano-González* [a]

[a] *Department of Cardiology, UMAE Hospital de Cardiologia, Centro Médico Nacional Siglo XXI, Mexican Social Security Institute, Mexico City, Mexico*

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ABSTRACT

Background: Congenital absence of the right coronary artery is an exceptionally rare coronary anomaly and represents an uncommon variant of a single coronary artery. Most patients remain asymptomatic, and the condition is often identified incidentally during cardiovascular imaging performed for unrelated clinical indications. Multimodality imaging plays a key role in establishing the diagnosis and accurately defining coronary anatomy.

Case Presentation: A 54-year-old man with a history of rheumatic heart disease, mechanical mitral valve replacement, permanent atrial fibrillation, heart failure with reduced ejection fraction, and cardiac resynchronization therapy defibrillator implantation was referred for preoperative cardiovascular evaluation due to prosthetic mitral valve dysfunction. Coronary angiography demonstrated the absence of a right coronary artery origin and a dominant left circumflex artery supplying the entire right coronary territory, without significant obstructive coronary artery disease. Coronary computed tomography angiography with three-dimensional reconstruction confirmed congenital absence of the right coronary artery and demonstrated an ultra-dominant left circumflex artery coursing along the right atrioventricular groove.

Conclusion: This case highlights the value of multimodality imaging, particularly the complementary use of coronary angiography and coronary computed tomography angiography, for the diagnosis, anatomical characterization, and differentiation of rare coronary artery anomalies from acquired coronary pathologies.

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* *Corresponding author.*
E-mail address: louis.roblesmi@gmail.com
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Ausencia congénita de la arteria coronaria derecha con arteria circunfleja ultradominante: diagnóstico por imagen multimodal

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RESUMEN

Antecedentes: La ausencia congénita de la arteria coronaria derecha es una anomalía coronaria excepcionalmente rara y representa una variante poco común de una arteria coronaria única. La mayoría de los pacientes permanecen asintomáticos y la afección suele identificarse de forma incidental durante estudios de imagen cardiovascular realizados por indicaciones clínicas no relacionadas. Las técnicas de imagen multimodal desempeñan un papel fundamental en el diagnóstico y la definición precisa de la anatomía coronaria.

Presentación del caso: Un hombre de 54 años con antecedentes de cardiopatía reumática, reemplazo valvular mitral mecánico, fibrilación auricular permanente, insuficiencia cardíaca con fracción de eyección reducida e implantación de un desfibrilador para terapia de resincronización cardíaca fue derivado para evaluación cardiovascular preoperatoria debido a disfunción de la prótesis valvular mitral. La angiografía coronaria demostró la ausencia del origen de la arteria coronaria derecha y una arteria circunfleja izquierda dominante que irrigaba todo el territorio coronario derecho, sin enfermedad coronaria obstructiva significativa. La angiografía coronaria por tomografía computarizada con reconstrucción tridimensional confirmó la ausencia congénita de la arteria coronaria derecha y demostró una arteria circunfleja izquierda ultradominante que discurría a lo largo del surco auriculoventricular derecho.

Conclusión: Este caso resalta el valor de las técnicas de imagen multimodal, en particular el uso complementario de la angiografía coronaria y la angiografía coronaria por tomografía computarizada, para el diagnóstico, la caracterización anatómica y la diferenciación de anomalías coronarias raras de patologías coronarias adquiridas.

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1. INTRODUCTION

Coronary artery anomalies (CAA) comprise a group of congenital malformations characterized by abnormal development involving the origin, course, and/or termination of the epicardial coronary arteries [1]. They have an overall incidence of <1% in the general population, and their clinical relevance lies in their association with adverse cardiovascular events [1].

Among CAA involving coronary origin, the presence of a single coronary artery (SCA) has an even lower estimated incidence, ranging from approximately 0.056–0.67%, with congenital absence of branches of the left main coronary artery (LMCA), such as congenital absence of the left circumflex artery (LCx), being more frequently reported [1, 2]. In contrast, congenital absence of the right coronary artery (RCA) has an even lower incidence, with only a few cases reported in the international literature. It has been described in association with other coronary anomalies or cardiovascular entities such as LMCA dominance or ultra-dominant LCx [2, 3], which explains why patients with this anomaly are frequently asymptomatic or diagnosed incidentally.

We present the case of a patient in whom congenital absence of the RCA associated with an ultra-dominant LCx was documented during preoperative cardiovascular evaluation by coronary angiography (CAG) and coronary computed tomography angiography (CCTA).

2. CASE REPORT

A 54-year-old male with a history of mechanical mitral valve replacement in 2007 due to rheumatic heart disease, permanent atrial fibrillation, heart failure with reduced ejection fraction, a left ventricular ejection fraction (LVEF) of 30%, and a cardiac resynchronization therapy defibrillator (CRT-D) was referred to our institution for the first time for specialized evaluation. No prior medical

records available at our tertiary care center.

He was admitted for preoperative evaluation due to progressive dyspnea of three months' duration, corresponding to New York Heart Association (NYHA) functional class II–III. Laboratory studies were essentially normal, with preserved renal function and suboptimal therapeutic anticoagulation (INR 1.98). Electrocardiography demonstrated atrial fibrillation with a mean ventricular response of 75 bpm and no other relevant findings. Transthoracic echocardiography revealed global left ventricular hypokinesia, LVEF of 30%, and dysfunction of the mechanical mitral prosthesis with a mean transmitral gradient of 6.6 mmHg and an effective valve area of 0.6 cm² calculated by the continuity equation. Transesophageal echocardiography confirmed immobility of one of the prosthetic hemidiscs, without evidence of thrombus, with findings suggestive of pannus (Figure 1). Due to the need for repeat valve intervention and the absence of available coronary anatomical information, preoperative evaluation was completed by cardiac catheterization.

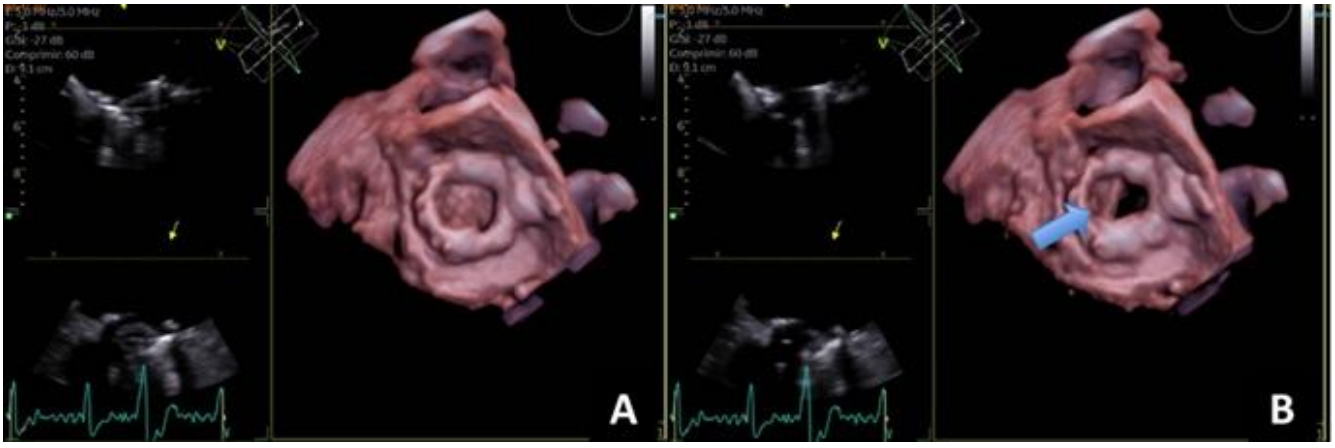


Figure 1: *Three-dimensional transesophageal echocardiography of the mechanical mitral prosthesis. (A) En-face atrial view showing restricted mobility of one prosthetic hemidisc. (B) Corresponding image demonstrating incomplete opening of the affected hemidisc (blue arrow), consistent with prosthetic valve dysfunction.*

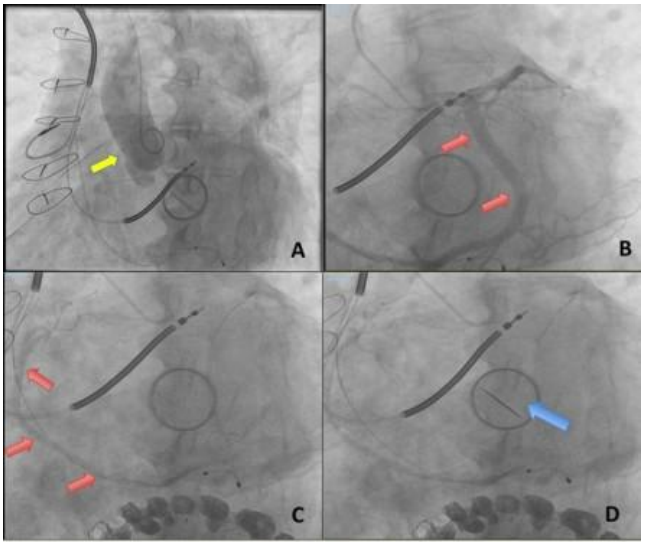


Figure 2: *Findings obtained during invasive cardiac catheterization. (A) Aortography demonstrates absence of a right coronary origin from the aortic root (yellow arrow). (B) Left coronary angiography showing a dominant circumflex artery (red arrows) without significant obstructive disease. (C) Distal circumflex artery extending beyond the crux (red arrows) and supplying the territory normally perfused by the right coronary artery. (D) Fluoroscopic evaluation of the mechanical mitral prosthesis demonstrating restricted mobility of one hemidisc (blue arrow), consistent with prosthetic valve dysfunction.*

CAG (Figure 2) demonstrated a left main coronary artery without angiographically significant lesions, a left anterior descending artery free of obstructive disease, and a dominant LCx. The RCA ostium could not be identified despite multiple attempts at selective cannulation. Given this finding, aortography was performed in multiple projections, confirming the absence of a right coronary origin from the aortic root. The LCx extended beyond the cardiac crux, giving rise to branches responsible for perfusion of the territory normally supplied by the RCA. No significant obstructive coronary lesions were identified.

The main differential diagnoses considered were chronic ostial occlusion of the RCA and anomalous coronary origin. In order to characterize the coronary anatomy, CCTA with three-dimensional reconstruction was performed (Figure 3). The study confirmed congenital absence of the RCA, identifying an LMCA and an ultra-dominant LCx coursing along the right atrioventricular groove and providing complete perfusion of the right coronary territory. Additionally, CCTA confirmed immobility of one hemidisc of the mechanical mitral prosthesis and findings compatible with pannus.

Integration of the findings obtained by CAG and CCTA established the definitive diagnosis of congenital absence of the right coronary artery with an ultra-dominant LCx. A multidisciplinary Heart Team discussed the case, and surgical mitral valve replacement was recommended due to the documented prosthetic valve dysfunction.



Figure 3: *Coronary computed tomography angiography confirming congenital absence of the right coronary artery. (A) Axial image showing contrast opacification of the left coronary system without an identifiable right coronary origin (blue arrow). (B) Three-dimensional reconstruction demonstrates a single left coronary system with an ultra-dominant circumflex artery (red arrow) extending along the right atrioventricular groove and providing complete perfusion of the right coronary territory.*

3. DISCUSSION

Coronary artery development takes place between the fourth and eighth weeks of gestation. During this period, a primitive coronary vascular plexus is formed from endothelial precursor cells derived from the sinus venosus and the endocardium, which subsequently undergoes a process of remodeling and maturation mediated by epicardium-derived cells [4, 5]. Alterations in these embryological mechanisms may result in congenital coronary anomalies, including the congenital absence of the RCA observed in our case.

SCA may originate from the right sinus of Valsalva in up to 49% of cases, from the left sinus of Valsalva in 45% of cases, and in the remaining cases from the pulmonary artery trunk or other anomalous sites [6].

Lipton et al. described an angiographic classification with clinical correlation for SCA cases, establishing three groups and their respective subgroups. When the SCA originates from the left sinus of Valsalva and gives rise to a left anterior descending artery with a normal course and trajectory, while the LCx is ultra-dominant, continuing beyond the crux into the right atrioventricular groove and giving rise to marginal branches supplying the right ventricle, it is classified as Lipton type L-I [7]. This classification corresponds to that observed in our patient by both CAG and CCTA.

The use of CCTA, due to its excellent spatial resolution, adequate tissue resolution, and ability to determine the anatomical relationships between cardiovascular structures and surrounding extracardiac structures, represents a necessary complementary imaging modality that correlates with the findings documented by CAG [8].

4. CONCLUSIONS

Our patient had an extensive cardiovascular history, but the absence of information regarding his coronary anatomy and the need to determine eligibility for mitral valve replacement surgery based on the documented findings warranted a comprehensive evaluation with CCTA, ultimately establishing the definitive diagnosis of SCA due to congenital absence of the RCA. In the absence of malignant coronary courses or significant atherosclerotic disease, as in this case, most patients remain asymptomatic and do not require specific treatment for the coronary anomaly itself.

5. CONFLICT OF INTERESTS

The authors have no conflict of interest to declare. The authors declared that this study has received no financial support.

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